

Author's Contributions

The Continuous-Coefficient Jensen Equation: How Real Mixing Weights Retire Regularity

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Conceptualization. A.E. conceived the central distinction between the discrete-coefficient and continuous-coefficient Jensen equations as a separable dictionary problem, identified the five classical regularity hypotheses that become vestigial under the continuous-coefficient equation, and formulated the diagnostic framework (Section 4 of the manuscript) for classifying saturated-Jensen derivations into four structurally distinct sources.

Formal results and proofs. A.E. formulated and proved all results in the manuscript: the main theorem (Theorem 1, endpoint substitution), its strict-minimum variant (Theorem 2), the extension to convex domains of arbitrary real vector spaces (Theorem 3), the derived regularity corollary (Corollary 1), the piecewise-saturation corollary (Corollary 2), the irrational-weights remark (Remark 1), the Hamel-pathology proposition for the rational-coefficient equation (Proposition 1), and the resolution-blindness characterization (Proposition 2).

Literature review and attribution. A.E. relied heavily on Aczél's corpus, Kuczma's corpus and bibliography, and Daniel Reem's 2017 bibliography when assembling the classical regularity-hypothesis literature from Cauchy (1821) through Kestelman (1947); he also assembled and verified the historical attribution table (Table 1 and the mirrored Table 2) and identified the structural parallel with Fischer's 1972 regularity-free entropy characterization for $n > 2$ outcomes.

Formal verification. The core lemma of Theorem 1 was initially discovered through a Lean 4 formalization project in which the boundedness hypothesis appeared unused; A.E. designed the formalization harness, interpreted the formal proof output, and drew the mathematical conclusion presented in the provenance note.

Figures and tables. A.E. designed and implemented all three manuscript figures in TikZ: the load diagram (Figure 1, illustrating the dictionary of vestigial struts), the mechanism diagram (Figure 2, illustrating the endpoint substitution and the weight axis), and the resolution-axis diagram (Figure 3, illustrating surrogate calibration and the tent representation). A.E. composed both tables.

Writing. A.E. wrote the manuscript in its entirety, including all sections, proofs, figure captions, backmatter, and bibliography, and revised it.

Submission preparation. A.E. assembled the submission package, including the full-author and anonymous manuscript sources, cover letter, and all supplementary assets, in accordance with the Monthly's double-blind submission policy.